The instability of negative-mass Schwarzschild is a point-mass effect:

Exact thresholds for finite stars and a radially stable negative-mass boson star

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Abstract

Negative-mass Schwarzschild spacetime is linearly unstable: Gleiser and Dotti exhibited, for every multipole $\ell \geq 2$, an even-parity mode growing as $e^{\gamma t}$ with $\gamma = (\ell-1)\ell(\ell+1)(\ell+2)/(12|M|)$, and gave numerical evidence that it is the only one. It has been suggested that the instability does not depend on the singularity and would survive in a regular negative-mass star. We prove, for every $\ell \geq 2$, that the Dotti–Gleiser operator has exactly one negative eigenvalue. The proof uses the zero-energy partner solution $\phi_0 = 1/\psi_0$ implicit in their intertwining construction, through a ground-state identity and Glazman's lemma. The same identity yields an exact count of unstable modes for the exterior of a star of areal radius R whose interior imposes a Robin condition $\hat{\Psi}' = \kappa\hat{\Psi}$: $\#\{\gamma > 0\} = [\, R < r_s \,] + [\, \kappa < \kappa_c(R) \,]$, with $r_s = 6|M|/[(\ell+2)(\ell-1)]$ and κ_c in closed form. Consequently the vacuum exterior of any star with $R \geq r_*(\ell)$ ($r_* = 2.279|M|$ for $\ell = 2$) and a non-energy-injecting surface condition ($\kappa \geq 0$) supports no unstable even-parity mode, and the ℓ^4 growth of the point mass is capped by any finite radius. We then study a concrete regular negative-mass object, a sextic boson star whose stress tensor enters Einstein's equations with a minus sign. We derive its radial perturbation equations, verify them symbolically, and find the configuration used in a recent full numerical-relativity study to be radially stable; the family loses stability exactly where the Noether charge is extremal. Because the first law becomes $\mathrm{d}M = -\omega\,\mathrm{d}N$, the turning-point criterion holds in N but reverses sign when written in M. Accurate spherically symmetric nonlinear evolutions confirm both the stationarity of the stable stars to $t = 1500$ and the predicted growth rates of unstable ones.

1 Introduction

Bondi's 1957 analysis [1] showed that general relativity contains no logical obstruction to negative mass, provided inertial, passive and active mass share a sign, and that a pair of equal and opposite masses self-accelerates (the "runaway"). Two questions have dominated the subsequent discussion: whether negative mass is dynamically stable, and whether the runaway is physical. The second was recently addressed by full $3+1$ evolutions of a canonical boson star paired with a "phantom" boson star of negative ADM mass [3]. Here we address the first.

For the vacuum point mass the answer is known. Gibbons, Hartnoll and Ishibashi [4] proved stability of the odd-parity sector and, for a range of boundary conditions at the naked singularity, of the even sector. Gleiser and Dotti [5] then found exact, smooth, exponentially growing even-parity solutions and showed that the stability argument of Ref. [4] relies on a perturbation that

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is not continuous at an interior radius where the Zerilli variable is singular. Dotti and Gleiser [6] removed that "kinematic" singularity with an intertwining operator, obtained a regular operator $\hat{H}$ with a unique self-adjoint extension, and showed that the unstable mode is excited by generic data supported away from the singularity. Cardoso and Cavaglià [7] identified the modes as algebraically special. Whether the unstable mode is the only one was left as numerical evidence [6].

What a regular negative-mass source does to this instability has not been computed. In a recent review Dotti suggests that replacing the singularity by a small negative-mass distribution would leave the potential largely unchanged, so that "an unstable mode could be found anyway", and that "the singularity plays no role in the instability" [8]. Regular negative-mass configurations exist—perfect-fluid bubbles in de Sitter [9], scalar-field stars [3]—but stability analyses have been radial and restricted to special matter models [10, 11].

This paper makes three contributions.

1. We prove that $\hat{H}$ has exactly one negative eigenvalue for every $\ell \geq 2$ (Theorem 1), turning the numerical observation of Ref. [6] into a theorem.

2. For the exterior of a star of radius R with a Robin interface condition, we give an exact count of unstable modes for every ℓ (Theorem 2). The instability is a property of the region $r \lesssim 10|M|/\ell^2$: a star larger than $r_*(\ell)$ hides it unless its surface condition injects energy, and any finite star caps the growth rate. This quantifies, and for dilute stars contradicts, the expectation of Ref. [8].

3. We give the first linear stability analysis of a negative-ADM-mass boson star [3]. It is radially stable on the branch used in numerical relativity, with stability lost exactly at the extremum of the Noether charge.

The non-radial stability of the boson star, which requires computing the effective interface condition that its interior supplies, is left for future work (Sec. 6).

Units $G = c = 1$. Sections 2–4 use $|M| = 1$; Sec. 5 uses the scalar mass $m = 1$. All computations are reproducible from the accompanying repository.

2 The intertwined Zerilli operator

Write the negative-mass Schwarzschild metric as $\mathrm{d}s^2 = -f\mathrm{d}t^2 + f^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2$ with $f = 1 - 2M/r = 1 + 2/r$ ($M = -1$), and the tortoise coordinate $x = r - 2\ln(1 + r/2)$, so $x \in (0, \infty)$ with $x \simeq r^2/4$ at the singularity. Even-parity perturbations with multipole $\ell \geq 2$ reduce to Zerilli's equation [13] $\partial_t^2 \Psi_Z + H_Z \Psi_Z = 0$, $H_Z = -\partial_x^2 + V_Z$. Let

$$c = (\ell + 2)(\ell - 1), \qquad k = \frac{(\ell - 1)\ell(\ell + 1)(\ell + 2)}{6} = \frac{c(c + 2)}{6}, \qquad r_s = \frac{6}{c}. \tag{1}$$

For $M < 0$ the potential V_Z has a double pole at r_s, which lies in the physical domain; it is "kinematic" because Ψ_Z itself has a simple pole there for smooth metric perturbations [5].

Dotti and Gleiser [6] use the static solution $\psi_0 = \chi(r)/(cr - 6)$ of $H_Z \psi_0 = 0$, where χ is the polynomial of degree $\ell + 2$ in their Eq. (55), and define $g = \psi_0'/\psi_0$ ($' = \mathrm{d}/\mathrm{d}x$),

$$A = \partial_x + g, \qquad H_Z = AA^*, \qquad \hat{H} = A^*A = -\partial_x^2 + \hat{V}, \qquad \hat{V} = g^2 - g'. \tag{2}$$

$\hat{V}$ is regular, $\hat{H}A^* = A^* H_Z$, and $\hat{\Psi} = A^* \Psi_Z$ is smooth for smooth metric perturbations. The zero-energy partner of this factorisation,

$$\phi_0 = \frac{1}{\psi_0} = \frac{cr - 6}{\chi(r)}, \qquad A\phi_0 = 0, \qquad \hat{H}\phi_0 = 0, \tag{3}$$

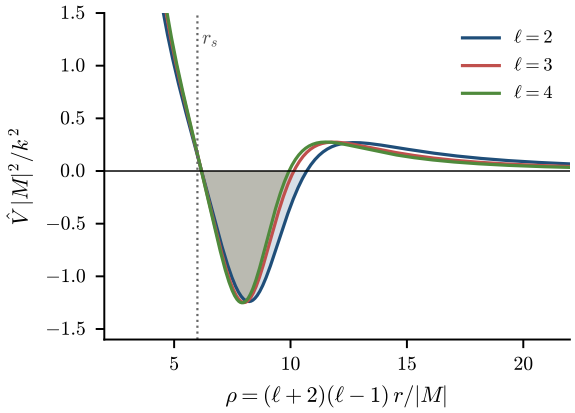


Figure 1: The regularised potential $\hat{V}$ for $\ell = 2, 3, 4$, in the scaled radius $\rho = cr/|M|$. Shaded: $\hat{V} < 0$, which occurs only on a shell just beyond r_s ($\rho = 6$). With radius scaled by $c = (\ell+2)(\ell-1)$ and the potential by the eigenvalue scale k^2, the curves nearly coincide.

appears in Ref. [6] as the state "missing in the spectrum" of the supersymmetric partner. It is the object on which everything below rests. Figure 1 shows $\hat{V}$: it is negative only on a thin shell just outside r_s, and when lengths are scaled by c and energies by k^2 the shell is nearly independent of ℓ.

Lemma 1 (properties of χ, ϕ_0, $\hat{V}$; all $\ell \geq 2$)**.**

1. $\chi = \sum_{n=1}^{\ell+2} c_n r^n$ *with* $c_1 = \frac{3}{2}$, $c_2 = 0$, $c_3 = -K/32$, $c_4 = K/96$ *($K = 6k$),* $c_n \geq 0$ *for* $n \geq 4$ *and* $c_{\ell+2} > 0$*;* ψ_0 *solves the static Zerilli equation identically in* ℓ*; and* $\chi > 0$ *on* $r > 0$*.*

2. *At* r_s *the* χ*-equation has Frobenius exponents* 0 *and* 3*, forcing* $\chi''(r_s) = 0$*; hence* $\hat{V}$ *is real-analytic on* $(0, \infty)$*.*

3. $r^4\hat{V} = 12 + O(r)$ *as* $r \to 0$ *(exponents* $x^{3/2}$*,* $x^{-1/2}$*), and* $\hat{V} \simeq (\ell+1)(\ell+2)/r^2$ *as* $r \to \infty$*.* $\hat{H}$ *is limit point at both ends, hence essentially self-adjoint on* $C_c^\infty(0, \infty)$*.*

4. ϕ_0 *is the non-admissible branch* $\propto x^{-1/2}$ *at* 0*, decays as* $r^{-(\ell+1)}$ *at infinity, and has exactly one zero, at* r_s*, which is simple.*

Sketch. (1) The closed form $c_n = (n-2)\big((n-4)\ell(\ell+1) + n - 1\big)(\ell - n + 3)_{2n-4}/\big(2^n(n-1)!^2\big)$ turns the static equation into a three-term recurrence, which becomes a polynomial identity in (ℓ, n) after dividing by a Pochhammer factor; it is verified symbolically. Positivity follows Ref. [6], Eqs. (57)–(59): dropping the non-negative terms $n \geq 6$ leaves a quartic whose unique positive critical value is $4(K-18)(Ks - 18(K-16))/[(\ell-2)(\ell+3)(\ell^2+\ell+4)]^3$, $s = \sqrt{6(K-18)}$, which is positive because $(Ks)^2 - (18(K-16))^2 = 6(K-24)^3 > 0$ for $\ell \geq 3$; $\ell = 2$ is checked directly. (2)–(4) are local expansions, verified symbolically in ℓ; limit point at 0 holds because $x^{-1/2} \notin L^2$, and the Weyl alternative gives essential self-adjointness [15, 16]. □

3 Exactly one unstable mode

The key tool is the ground-state identity: if $\hat{V} = \phi_0''/\phi_0$ and $u = \phi_0 h$, then

$$u'^2 + \hat{V}u^2 = \phi_0^2 h'^2 + \big(\phi_0\phi_0' h^2\big)'. \tag{4}$$

Theorem 1. *For every $\ell \geq 2$, $\hat{H}$ has exactly one negative eigenvalue. It is simple and equals $-(k/2)^2$, so the unique unstable even-parity mode of negative-mass Schwarzschild grows at the Gleiser–Dotti rate $\gamma = k/(2|M|)$. Zero is not an eigenvalue.*

Proof. At most one. By Glazman's lemma [17], the number of negative eigenvalues equals the largest dimension of a subspace of a form core on which $q[u] = \langle u, \hat{H}u\rangle$ is negative definite. Take the core $C_c^\infty(0,\infty)$ and suppose such a subspace F has $\dim F \geq 2$. Choose $u \in F \setminus \{0\}$ with $u(x_s) = 0$, $x_s = x(r_s)$. Since ϕ_0 is analytic with a simple zero at x_s, $h = u/\phi_0$ is smooth, and integrating (4) over the compact support of u gives $q[u] = \int \phi_0^2 h'^2 \,\mathrm{d}x \geq 0$, a contradiction.

At least one. The function $\hat{\Psi} = A^*\Psi_{\mathrm{DG}}$, with $\Psi_{\mathrm{DG}} = r(r+2)^k e^{-kr/2}/(cr-6)$, satisfies $\hat{H}\hat{\Psi} = -(k/2)^2\hat{\Psi}$ identically in ℓ, is $O(r^3)$ at 0, regular at r_s, and decays exponentially; so it is an eigenfunction. Independently of this explicit mode, $w = \phi_0 \mathbf{1}_{x>x_s}$ lies in the form domain with $q[w] = 0$ by (4); if $\hat{H} \geq 0$, w would be a minimiser and hence a weak, therefore C^1, solution of $\hat{H}w = 0$, contradicting the kink at x_s.

No zero mode. The zero-energy solution admissible at $x = 0$ is

$$v = \phi_0 F, \qquad F(r) = \text{f.p.}\int_0^r \psi_0^2 \,\frac{\mathrm{d}x}{\mathrm{d}r}\,\mathrm{d}r;$$

it grows as $r^{\ell+2}$ and is not in L^2. □

Remark 1. v has exactly one zero for every ℓ (F is positive on $(0, r_s)$ and increases from $-\infty$ to $+\infty$ on (r_s, ∞)), as renormalised oscillation theory [18] requires. The proof does not use oscillation theory at the continuum edge. Numerically, Chebyshev discretisations with $\hat{\Psi} = r^3 u$ reproduce $E_0 = -(k/2)^2$ to $\leq 3 \times 10^{-13}$ for $\ell \leq 8$ and find no second bound state in 140 resolution and domain variations.

4 Finite stars: an exact count

Model the exterior of a star of areal radius R by $\hat{H}_{R,\kappa}$ on $L^2(x_R, \infty)$ with the Robin condition $\hat{\Psi}'(x_R) = \kappa\hat{\Psi}(x_R)$; $\kappa = +\infty$ denotes Dirichlet. The interior matter determines κ, which in general depends on frequency and ℓ; here it is a free parameter. Define

$$\kappa_c(R) = \frac{\mathrm{d}\ln\phi_0}{\mathrm{d}x}\Big|_R = \Big(1 + \frac{2}{R}\Big)\frac{\mathcal{N}(R)}{(cR-6)\chi(R)}, \qquad \mathcal{N} = c\chi - (cr-6)\chi'. \tag{5}$$

Theorem 2. *For every $\ell \geq 2$ and $R \neq r_s$, the number of negative eigenvalues of $\hat{H}_{R,\kappa}$ is*

$$[\,R < r_s\,] + [\,\kappa < \kappa_c(R)\,], \tag{6}$$

with Dirichlet counted as $\kappa = +\infty$. At $\kappa = \kappa_c$, ϕ_0 is a zero-energy eigenfunction.

Proof. For u in the core, (4) gives $q[u] = \int \phi_0^2 h'^2 \,\mathrm{d}x + (\kappa - \kappa_c)u(x_R)^2$ provided $u(x_s) = 0$ when $R < r_s$. Imposing $u(x_s) = 0$ (if $R < r_s$) and $u(x_R) = 0$ (if $\kappa < \kappa_c$) costs at most $[\,R < r_s\,] + [\,\kappa < \kappa_c\,]$ linear conditions and leaves $q \geq 0$; Glazman's lemma gives the upper bound. For the lower bound use $w_1 = \phi_0 \mathbf{1}_{[x_R, x_s]}$ and $w_2 = \phi_0 \mathbf{1}_{[x_s,\infty)}$ (or ϕ_0 itself if $R > r_s$), for which $q[aw_1 + bw_2] = a^2\phi_0(x_R)^2(\kappa - \kappa_c)$ and $q[w_2] = 0$. If the negative spectral subspace were smaller, some combination orthogonal to it would have $q = 0$, hence be a weak zero-energy solution satisfying the boundary condition; the kink of w_2, or $\kappa \neq \kappa_c$, excludes this. □

ℓ	r_s	r_*	$r_{\rm neg}$	$\gamma_{\rm DG} = k/2$	$\lvert M\rvert/r_*$
2	1.5000	2.2788	2.6741	2	0.439
3	0.6000	0.8789	1.0143	10	1.14
4	0.3333	0.4814	0.5519	30	2.08
5	0.2143	0.3073	0.3512	70	3.25

Table 1: Thresholds in units of $|M|$. r_s: Dirichlet threshold; r_*: Neumann and $\kappa \geq 0$ threshold; $r_{\rm neg} = \sup\{r : \hat{V} < 0\}$. Since r_* decreases with ℓ (checked for $\ell \leq 80$), a star with $|M|/R < 0.439$ and $\kappa \geq 0$ has no unstable even-parity mode for any ℓ. At large ℓ, $r_* \approx 8.486/c$ and $r_{\rm neg} \approx 9.636/c$ (derived from the $\ell \to \infty$ limit of $c\,\chi(\rho/c)$, a modified-Bessel closed form; convergence of the roots is not proven).

Corollary 1 (thresholds, all ℓ)**.** *$\mathcal{N} > 0$ on $[0, r_s]$ and has exactly one root $r_* > r_s$. Hence $\kappa_c < 0$ on $(0, r_s)$, $\kappa_c \to \mp\infty$ at $r_s^{\mp}$, $\kappa_c > 0$ on (r_s, r_*) and $\kappa_c < 0$ on (r_*, ∞). In particular: Dirichlet is unstable iff $R < r_s$; Neumann is unstable iff $R < r_*$; every $\kappa \geq 0$ is stable iff $R \geq r_*$.*

The first claim follows from a telescoping lower bound on $[0, r_s]$; uniqueness of r_* from Descartes' rule applied to $\mathcal{N}(r_s+t)$, whose coefficients have one sign change because $\chi''(r_s) = 0$ and $\chi^{(m)}(r_s) > 0$ for $m \geq 4$. Table 1 lists the thresholds and Fig. 2 the critical surface condition. Asymptotically,

$$\kappa_c = -\frac{\ell+1}{R} - \frac{\ell^4 + 3\ell^3 + 4\ell^2 + 4}{\ell(\ell-1)(\ell+2)}\frac{1}{R^2} + O(R^{-3}) \qquad (R \to \infty), \tag{7}$$

$$\kappa_c\, x_R = -\frac{1}{2} - \frac{(\ell^2+\ell-1)R}{12} + O(R^2) \qquad (R \to 0). \tag{8}$$

For large R an instability therefore requires a surface condition softer than the static vacuum tail $\phi_0 \propto r^{-(\ell+1)}$ by an $O(1)$ factor on the scale $1/R$. As $R \to 0$ every condition with $\kappa x_R > -\frac{1}{2}$ (including Dirichlet and Neumann) reproduces the Dotti–Gleiser eigenvalue, with error scaling roughly as R^3–R^4, so the self-adjoint extension of Ref. [6] is the limit of a regular cutoff; compare the scalar-wave analysis of Ref. [14].

Growth-rate cap. For the point mass the growth rate $k/2 \sim \ell^4/12$ is unbounded in ℓ. For a star: with $\kappa \geq 0$ only ℓ with $R < r_*(\ell)$, roughly $\ell \lesssim 3/\sqrt{R}$, are unstable; with a fixed $\kappa < 0$, every ℓ with $r_{\rm neg}(\ell) < R$ has $\hat{V} \geq 0$ outside the star and growth $\leq |\kappa|$. Either way $\sup_\ell \gamma < \infty$ (Fig. 2, right; a 1000-case numerical scan agrees with Theorem 2). The unbounded growth belongs to the singularity.

5 A regular negative-mass star

Theorem 2 reduces the even-parity stability of any star with $R > r_*$ to the sign of the effective κ its interior supplies, together with the stability of the matter itself. We now examine the matter for a concrete model.

Model. Following Ref. [3], take a complex scalar with $\mathcal{L} = -\frac{1}{2}|\nabla\Phi|^2 - V(|\Phi|^2)$, $V(s) = \frac{1}{2}s - \frac{1}{4}\lambda s^2 + \frac{1}{6}\mu s^3$, $\lambda = 10240$, $\lambda^2/\mu = 4.8$, coupled through

$$G_{ab} = 8\pi\varepsilon\, T_{ab}[\Phi], \qquad \Box\Phi = 2V'(|\Phi|^2)\Phi, \tag{9}$$

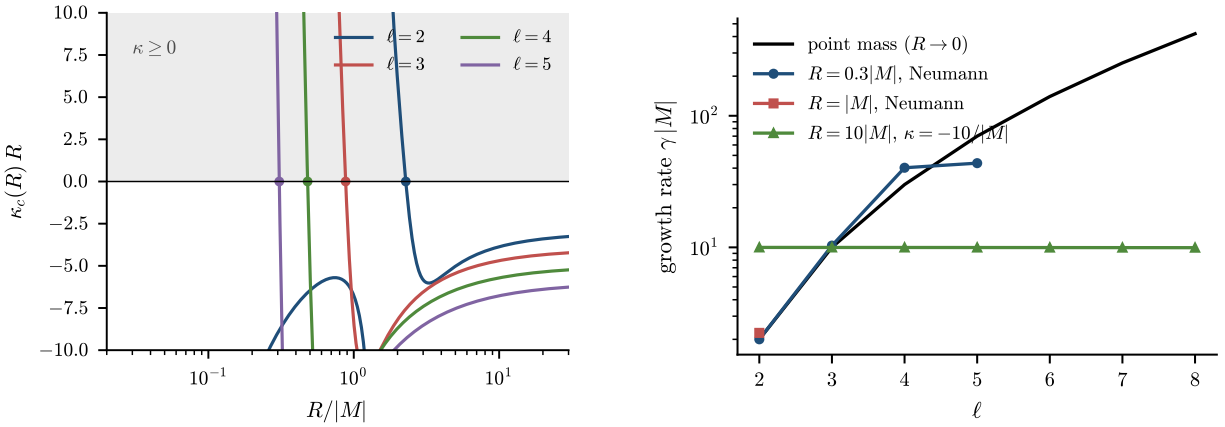


Figure 2: Left: critical surface condition $\kappa_c(R)R$ for $\ell = 2$–5. A mode is unstable when $\kappa < \kappa_c$, plus one unconditional mode when $R < r_s$ (pole). Dots mark r_*. Above r_* the whole $\kappa \geq 0$ region (shaded) is stable. Right: growth rate versus ℓ. The point mass grows as $\ell^4/12$; any finite R with an ℓ-independent condition leaves finitely many unstable ℓ ($\kappa \geq 0$) or caps the rate at $|\kappa|$ ($\kappa < 0$, since $\hat{V} \geq 0$ outside $r_{\rm neg}$).

with $\varepsilon = +1$ (canonical) or $\varepsilon = -1$. For $\varepsilon = -1$ the field equation is unchanged but the star gravitates with negative ADM mass. In the language of Bondi, the scalar has a single mass parameter that is inertial, passive and active at once, and is negative for $\varepsilon = -1$; gravity alone cannot distinguish this from Bondi's "all three negative" [1, 2]. The action is that of a ghost, so the model is a classical probe; quantum mechanically it is unstable to vacuum decay [19].

Static stars. Stationary solutions $\Phi = \phi(r)e^{-i\omega t}$ in Schwarzschild-like coordinates are solved as two-point boundary-value problems with a decaying Yukawa tail. The configuration of Ref. [3] at $\omega = 0.7603$ has $M = -0.014580$, rms radius 4.40 and $|M|/R \approx 3 \times 10^{-3}$, far below the threshold $|M|/R < 0.439$ of Table 1: the shell where negative-mass Schwarzschild would be unstable lies deep inside the matter. (Our masses are ADM masses; those quoted in Ref. [3] are the mass enclosed where ϕ falls to 5% of its central value and are 1.5–2.3% smaller; the field profiles agree to 10^{-5}.)

First law. With $N = \int 4\pi r^2 \omega \phi^2 (a/\alpha)\,\mathrm{d}r$ the Noether charge, the family satisfies

$$\mathrm{d}M = \varepsilon\,\omega\,\mathrm{d}N, \tag{10}$$

verified by finite differences to $< 10^{-4}$. Hence $M(\omega)$ and $N(\omega)$ are extremal at the same ω for either sign, but $\mathrm{d}M/\mathrm{d}\omega$ and $\mathrm{d}N/\mathrm{d}\omega$ have opposite signs when $\varepsilon = -1$.

Radial perturbations. Perturbing the metric functions and both components of the field with $e^{i\sigma t}$, we eliminate the metric through the tr and rr constraints and obtain a closed fourth-order system for the real perturbation, a gauge-invariant phase and the perturbed enclosed charge, containing σ only as σ^2 and ε symbolically. Substituting back, all six linearised field equations are satisfied identically. The structure is that of Refs. [20, 21]. Eigenvalues are found by compound-matrix shooting, which keeps the two-dimensional space of decaying solutions well conditioned, and are cross-checked by an independent finite-difference pencil that also returns complex eigenvalues.

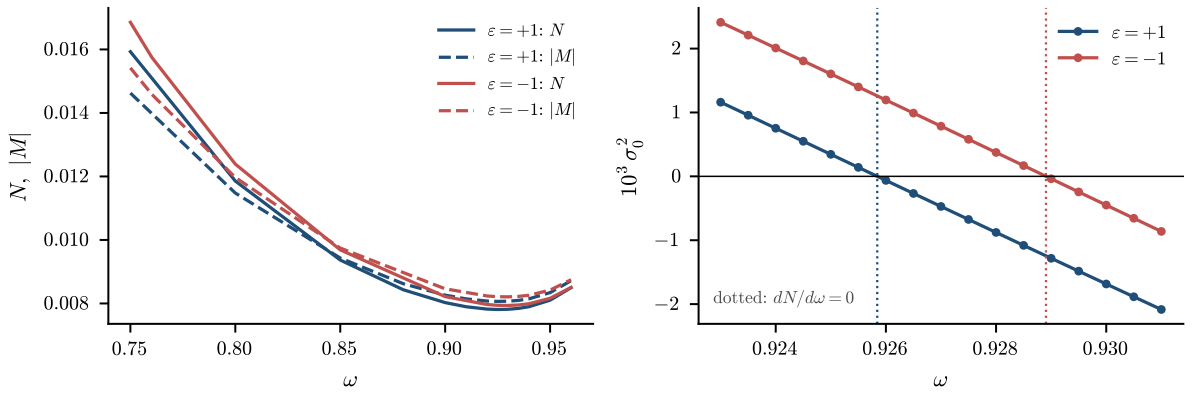


Figure 3: Radial stability of signed-source boson stars. Left: Noether charge N and $|M|$ along the canonical ($\varepsilon = +1$) and phantom ($\varepsilon = -1$) families. Right: fundamental σ_0^2 near the stability change; dotted lines mark $\mathrm{d}N/\mathrm{d}\omega = 0$, which coincides with $\sigma_0^2 = 0$ to 2×10^{-8} ($\varepsilon = +1$) and 8×10^{-8} ($\varepsilon = -1$) in ω.

Results. For $\varepsilon = +1$ the textbook criterion is reproduced at both turning points of the family: the fundamental mode crosses zero at $\omega = 0.92584444$ against $\mathrm{d}M/\mathrm{d}\omega = \mathrm{d}N/\mathrm{d}\omega = 0$ at 0.92584446, and again at the strong-field maximum of N ($\omega \approx 0.200$, $N \approx 42.73$) within the scan resolution. For $\varepsilon = -1$ (Fig. 3):

- The star of Ref. [3] is radially stable: no real growth rate up to $|\sigma| = 2$ and no complex eigenvalue with $|\sigma^2| < 0.3$, at two resolutions of the finite-difference check.
- The family is stable for $\omega < 0.92891$ and unstable above. The crossing $\sigma_0^2 = 0$ at $\omega = 0.92890846$ coincides with $\mathrm{d}N/\mathrm{d}\omega = 0$ at 0.92890838.
- The criterion is therefore invariant when stated in N (stable side $\mathrm{d}N/\mathrm{d}\omega < 0$) but reverses when stated in M: on the stable side $\mathrm{d}M/\mathrm{d}\omega = -\omega\,\mathrm{d}N/\mathrm{d}\omega > 0$.
- The phantom family has a fold in ω ($\omega_{\min} \approx 0.4223$) at which neither M nor N is extremal; the fundamental mode stays positive through it. The strong-field branch, followed to $|2m/r| \approx 0.4$, is stable with no turning point.

Near $\omega \approx 0.93$ the instability is essentially the flat-space Vakhitov–Kolokolov instability of Q-balls [22, 23], with gravity a small correction; the strong-field branch is the more sensitive test of how ε enters. The first law and the N-form of the criterion follow from the standard turning-point arguments [24–26] with ε kept; the new content is the explicit linear analysis for $\varepsilon = -1$. The result is consistent with the theorem that barotropic negative-mass stars are radially unstable [11], which does not apply to scalar-field matter.

Nonlinear check. We evolve single stars in spherical symmetry in polar-areal slicing, re-solving the Hamiltonian constraint for the Misner–Sharp mass [27] and the slicing condition at every substep, with ε appearing only there [28, 29]. The scheme is fourth order in space and time and converges at order ≈ 4; the Misner–Sharp mass outside the star equals the ADM mass of the static solution to 5×10^{-8}. Unstable stars at $\omega = 0.94$, perturbed by $\pm\delta$, grow at $\sigma = 0.07376$ ($\varepsilon = +1$) and 0.06679 ($\varepsilon = -1$) against 0.07370 and 0.06668 from the linear analysis (Fig. 4, right). The

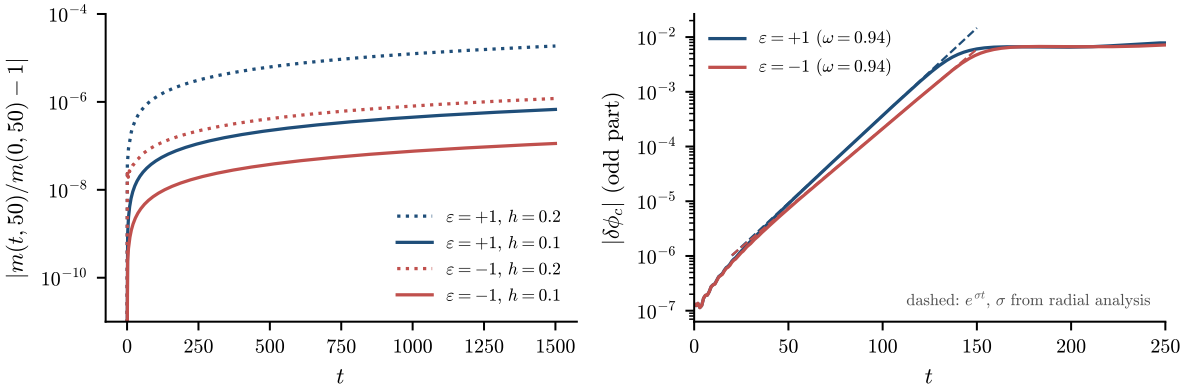


Figure 4: Spherically symmetric nonlinear evolutions. Left: relative drift of the Misner–Sharp mass at $r = 50$ for the stable campaign stars ($\omega = 0.750$ canonical, $\omega = 0.7603$ phantom) at two resolutions; the drift is a dissipation-limited truncation error that decreases rapidly with h. Right: odd part of the central-amplitude response of unstable stars ($\omega = 0.94$) to $\pm 10^{-5}$ perturbations; dashed lines have the growth rates of the linear analysis.

stable campaign stars of both signs remain stationary to $t = 1500$, with fractional drifts of ω, the central amplitude, the Misner–Sharp mass and N all $\leq 10^{-6}$ at $h = 0.1$ and decreasing with resolution (Fig. 4, left). A 5% perturbation relaxes within ~ 150 time units onto the neighbouring static star and stays there. The slow drifts reported for a lone phantom star in 3+1 [3] are therefore not a property of the spherical dynamics.

6 Discussion

For the point mass, Theorem 1 completes the linear stability analysis of negative-mass Schwarzschild: one unstable even-parity mode per ℓ, at the Gleiser–Dotti rate, and no others. Theorem 2 shows where that mode lives. The negative part of the effective potential sits on a shell of radius $\approx 10|M|/\ell^2$ (Fig. 1); a star whose surface lies outside $r_*(\ell)$ hides it unless its interior supplies an energy-injecting surface condition, and for $\ell = 2$ the condition is $|M|/R < 0.439$. In this sense the instability is a point-mass effect, contrary to the suggestion in Ref. [8]: for a dilute regular source the vacuum exterior cannot drive it, and the ℓ^4 growth disappears.

What remains is a property of the matter. For the boson star of Ref. [3], which is dilute ($|M|/R \sim 10^{-3}$), we have shown radial stability. Non-radial stability requires the effective interface condition $\kappa_{\rm eff}(\ell, \omega)$ produced by the interior, together with any instability intrinsic to the matter; the polar perturbation equations of the signed-source boson star are the natural route, and we leave them for future work. Theorem 2 reduces the gravitational part of that question to a comparison of $\kappa_{\rm eff}$ with the closed-form $\kappa_c(R)$.

Two limitations should be stated. Our results are classical: the phantom sector is a ghost, and a quantum theory of it has an unstable vacuum [19]. And the Robin model of the interior is exact only for an interface condition independent of frequency; a real star supplies a frequency-dependent one, for which Theorem 2 gives the count at each fixed frequency.

7 How the results were obtained

This work was carried out by the author working with an AI research assistant (Anthropic's Claude, run as an agent with a Python environment and the ability to delegate independent tasks to parallel sub-agents). The author set the questions, the scope and the priorities, and reviewed each stage; the assistant carried out the literature reading, derivations, code and drafting under that direction. Because AI-generated mathematics can be wrong in ways that read fluently, the workflow was built so that no claim rests on the assistant's word alone.

From literature to a list of gaps. We began from Bondi's paper [1] and read forward through the work it prompted [2, 4–11] and the recent simulation of Ref. [3]. For each paper we recorded what is proved, what is assumed and what is only asserted, and collected the open points into a list of gaps: whether the three Bondi masses can differ consistently, whether the runaway survives beyond Newtonian order, whether the known instability counts all unstable modes, whether it survives replacing the singularity by a star, and what the positive-mass theorem does and does not exclude. Each gap became a phase of a written plan, with a stated test for what would count as closing it.

Checking the starting points. Before building on any published result we re-derived it. The Zerilli potential, the Dotti–Gleiser intertwiner and the Gleiser–Dotti growth rate were reproduced symbolically; this exposed a sign misprint in one intermediate expression of Ref. [6], which we corrected before use, and the discontinuity of the truncated mode used in Ref. [4]. The static boson stars of Ref. [3] were reproduced to 10^{-5} in the field profile, and the 2% mass offset traced to the definition of mass used there.

Analytic results. The proofs of Lemma 1 and Theorems 1–2 were first written out as conventional arguments; then every algebraic identity on which they depend (the ground-state identity, the closed form of κ_c, the factorisations and sign properties of χ and $\mathcal{N}$) was verified in exact arithmetic with SymPy, for $\ell = 2$–10 individually and symbolically in ℓ where the text says so. Sign and root claims were checked by exact root isolation for $\ell \leq 50$. Each proof was then tested numerically: the predicted eigenvalue count was compared with direct spectral computations over a 1000-case grid in (ℓ, R, κ), with no disagreement.

Numerical results. Every eigenvalue reported in Sec. 5 was computed by two independent methods (compound-matrix shooting and a finite-difference generalised eigenproblem), and the stability changes were compared against the turning points of the families computed separately from the background solutions. The nonlinear evolution code was validated by its convergence order, by conservation of the Misner–Sharp mass, and by recovering the linear growth rates of unstable stars to 0.2%. Numerical codes use NumPy, SciPy and mpmath; all scripts and their recorded outputs are in the repository.

Independent review and novelty. Separate agents, given the code but not the conclusions, reviewed the public code accompanying Ref. [3] and our own codes for errors. A literature search was run for each result before calling it new; it showed, for example, that the first post-Newtonian separation law for a Bondi pair that we rederived had already been obtained [12], which is why it is not presented here, and that the zero mode $\phi_0 = 1/\psi_0$ on which Sec. 3 builds already appears in Ref. [6], as credited there.

Limitations of the process. The checks above are internal. The results have not yet been reviewed by specialists in the field, and the symbolic and numerical verifications establish the stated identities and spectra but cannot rule out an error in how a statement was set up. We would welcome independent scrutiny, and the repository is organised so that each claim can be traced to the script that checks it.

Reproducibility. The symbolic verifications (Lemma 1, Theorems 1–2, exact for $\ell = 2$–10 and symbolically in ℓ where stated; root isolation of $\mathcal{N}$ for $\ell \leq 50$), the spectral computations, the boson-star background and perturbation codes, the nonlinear evolution code, and the script that generates every figure are in the accompanying repository.

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